Additional Table 2 Characteristics of studies on trehalose intervention for animal models of tau pathology

Studies, country	Blinding of assessor	Intervention(s), control and setting (intervention schedule)	Animal characteristics (neurodegenerative diseases)	Outcome measure(s)	Clinical outcome(s)	Physiological change(s)	Adverse event(s)	Conclusion

Du et al. (2013) [1] China	Yes	**Intervention:** Trehalose (0.5%, injected into right lateral ventricles) **Control[1]:** Sucrose (0.5%, injected into right lateral ventricles) **Control[2]:** NaCl. (0.9%, injected into right lateral ventricles) ***WT:** None	**APP/PS1 transgenic mice (AD)** Intervention: $n = 4$ Control[1]: $n = 4$ Control[2]: $n = 4$ **WT** $n = 4$ **Background** C57BL/6J **Age (mon)** 8 **Gender** N/R	<u>Clinical</u> Morris water maze test - Spatial memory and learning <u>Biochemical</u> Level of plague - Aβ	Trehalose improved spatial memory and learning in transgenic mice.	Trehalose decreased level of Aβ plaque in the hippocampus of transgenic mice.	N/R	Trehalose improved impaired cognitive function in AD by decreasing level of protein aggregates in the hippocampus.
Lee et al. (2021) [2] Taiwan	N/R	**Intervention[1]:** Trehalose pre-treatment (0.5–1%, oral administration, 1 mon) **Intervention[2]:** Lactulose pre-treatment (0.5–1%, oral administration, 1 mon) **Control:** None	**Oligomeric $A\beta_{25-35}$ induced transgenic mice [AD]** Intervention[1]: $n = 15$ Intervention[2]: $n = 15$ Control: $n = 15$ **WT** Intervention[1]: $n = 15$ Intervention[2]: $n = 15$ Control: $n = 15$ **Background** C57BL/6J **Age (mon)**	<u>Clinical</u> Morris water maze test - Spatial memory and learning Y-maze - STM, motor activity, and anxiety Open field test - Locomotion and anxiety <u>Biochemical</u> Expression of synapses - Presynaptic synaptophysin - Postsynaptic PSD95	Both lactulose and trehalose attenuated short-term memory and learning retrieval deficits in transgenic mice.	Lactulose but not trehalose ameliorated reduction of synaptophysin and PSD95 in transgenic mice ↓ $GFAP^+$ in transgenic mice by lactulose and trehalose. Lactulose and trehalose ameliorated reduction of LC3II/LC3I ratio in transgenic mice. ↓ p62 in transgenic mice by lactulose and trehalose.	N/R	Lactulose and trehalose pre-treatment ameliorated cognitive impairment in AD by decreasing the level of protein aggregation via autophagy flux, reduction of reactive gliosis, and acted as a chaperone for protein refolding, with lactulose additionally enhanced the synaptic protein expression.

			2–3 **Gender** 90 Male	Expression of astrogliosis - GFAP^{+} Expression of autophagy marker - LC3II/LC3I ratio - p62 Expression of chaperones - HSP70 - Lamp2a		Lactulose and trehalose ameliorated reduction of HSP70 and Lamp2a in transgenic mice.		
Liu et al. (2020) [3] China	N/R	**Intervention:** Trehalose (2%, oral administration, 1 mon) **Control:** None	**APP23 transgenic mice [AD]** Intervention: *n* = 8 Control: *n* = 7 **Background** N/R **Age (mon)** 9 **Gender** N/R	<u>Clinical</u> N/R <u>Biochemical</u> Level of plagues - Aβ Protein levels - APP - BACE1 - PS1 - CTF	N/R	Trehalose decreased level of Aβ plaque by increasing APP and CTF in transgenic mice.	N/R	Trehalose decreased the level of protein aggregates by modifying the processing of APP.
Perucho et al. (2012) [4] Spain	N/R	**Intervention:** Trehalose (1%, oral administration, 2x/wk, 2 wk) + Isoflurane anaesthesia. [1%, 3 x, 1.5 h each time] **Control[1]** Isoflurane anaesthesia (1%, 3 x, 1.5 h each	**APP$_{swe}$ (Tg2576) transgenic mice [AD]** Intervention: *n* = 6 Control[1]: *n* = 6 Control[2]: *n* = 6 **WT** Intervention: *n* = 6 Control[1]: *n* = 6 Control[2]: *n* = 6 **Background:** C57BL6	<u>Clinical</u> Actimeter - Motor & anxiety-related activity Y-maze - STM, motor activity, and anxiety Rota-rod test - Motor coordination <u>Biochemical</u> Level of plagues	Co-treatment of isoflurane and trehalose improved STM, anxiety, and motor activity in transgenic mice.	Co-treatment of isoflurane and trehalose in transgenic mice: ↓ Isoflurane-induced number of TUNEL cells in hippocampus and cerebral cortex and enhanced expression of GFAP^{+}. ↑ Isoflurane-induces of HSP70 and CHIP; as well as LC3II/LC3I ratio and Beclin-1/β-actin.	N/R	Repetitive exposure to isoflurane confers pro-amyloidogenic property which can be prevented by trehalose. Trehalose improved motor and cognitive function in AD by decreasing level of protein aggregates through activation of autophagy, attenuation of

	time) **Control**[2] None.	**Age (mon)** 11 **Gender** 36 Female	- Aβ - tau - Phospho-tau Degree of apoptosis - TUNEL cells Expression of astrogliosis - $GFAP^{+}$ Expression of chaperones - HSP70 - CHIP Expression of autophagy marker - LC3II/LC3I ratio - Beclin-1/β-actin	↓ Isoflurane-induced levels of Aβ, tau, and phospho-tau plagues.	astrogliosis, and as a chaperone for protein folding.

Portbury et al. (2017) [5] Australia	N/R	**Intervention:** Trehalose (2%, 0.1 mL/10 g bodyweight, oral administration, daily, 31 d] **Control:** Vehicle (2% NaCl, 0.5% Na-carboxymethylcellulose, 0.5% benzyl alcohol, and 0.4% Tween 80, oral administration, daily, 31 d)	**APP (Tg2576) transgenic mice [AD]** Intervention: *n* = 15 Control: *n* = 12 **WT** Intervention: *n* = 10 Control: *n* = 10 **Background** N/R **Age (mon)** Intervention (Transgenic): 16.2 ± 3; (WT): 16.4 ± 3 Control (Transgenic): 16 ± 3; (WT): 16.2 ± 3 **Gender** N/R	<u>Clinical</u> Morris water maze test - Spatial memory and learning <u>Biochemical</u> Expression of synapses - Synaptophysin Expression of neurogenesis marker - Doublecortin Expression of growth factor - PGRN Metal analysis - Iron - Zinc - Copper Level of plagues - Aβ Expression of autophagy marker - LC3II/LC3I ratio	Trehalose improved spatial memory and learning in transgenic mice.	Trehalose increased levels of synaptophysin, doublecortin, and PGRN in the hippocampus and cortex of transgenic mice.	N/R	Trehalose improved cognitive function in AD by activation of neurogenesis and growth factor. Trehalose had no effect on autophagy and biometals responsible for cognitive function.
Pupyshev et al. (2022) [6] Russia	N/R	**Intervention[1]:** Trehalose (2%, oral administration, daily, 14 d] + Rapamycin [10 mg/kg, intraperitoneal injection, every 2 d, 7 d] + 3-MA [15 mg/kg,	**Oligomeric $A\beta_{25-35}$ induced transgenic mice [AD]** Intervention[1]: *n* = 10-17 Intervention[2]: *n* = 10-17 Intervention[3]: *n* = 10-17 Control[1]: *n* = 10-17 Control[2]: *n* = 10-17 Control[3]: *n* = 10-17	<u>Clinical</u> Passive avoidance test - Memory & learning Open field test - Locomotion and anxiety Plus-maze test - Anxiety <u>Biochemical</u> Expression of autophagy marker	Experimental groups did not differ on locomotion. Trehalose, rapamycin, and combination improved memory and learning in transgenic mice.	Trehalose and rapamycin increased LC3-II and gene expressions and decreased IBA1 in frontal cortex, hippocampus, and amygdala of transgenic mice; greater effect was observed with co-administration. 3-MA inhibited LC3-	N/R	Trehalose improved cognitive function in AD through decreasing level of protein aggregates by activating autophagy and suppressing reactive gliosis. Additive effect was observed in co-administration with rapamycin in biochemical but not

	intraperitoneal injection, every 2 d, 7 d] + $A\beta_{25-35}$ (injection, daily, 4 d) **Intervention[2]:** Trehalose. [2%, oral administration, daily, 14 d] + Rapamycin (10 mg/kg, intraperitoneal injection, every 2 d, 7 d) + $A\beta_{25-35}$ (injection, daily, 4 d] **Intervention[3]:** Trehalose [2%, oral administration, daily, 7 d] + $A\beta_{25-35}$ (injection, daily, 4 d) **Control[1]:** Rapamycin (10 mg/kg, intraperitoneal injection, every 2 d, 7 d) + $A\beta_{25-35}$ [injection, daily, 4	**WT** Control[4]: *n* = 10-17 **Background** C57BL6J **Age (mon)** 2 **Gender** Male	- LC3II - Gene of Beclin-1 - Gene of ligase Parkin - Gene of LC3II Expression of microgliosis - IBA1 Level of plague - Aβ <u>Brain atrophy</u> Neuronal density - Frontal lobe - Hippocampus - Amygdala	II in mice. Trehalose and rapamycin reduced Aβ in transgenic mice; no greater effect was observed with co-administration.	clinical performances.

		d] **Control[2]:** 3-MA [15 mg/kg, intraperitoneal injection, every 2 d, 7 d] + $A\beta_{25-35}$ (injection, daily, 4 d) **Control[3]:** $A\beta_{25-35}$ (injection, daily, 4 d) **Control[4]:** Saline (injection, daily, 4 d)					
Holler et al. (2016) [7] United States	N/R	**Intervention:** Trehalose (2%, oral administration, daily, 65 d) **Control[1]:** Sucrose (2%, oral administration, daily, 65 d] **Control[2]:** Vehicle (normal drinking water)	***Grn*$^{+/-}$ transgenic mice** **[FTD]** Intervention: *n* = 7 Control[1]: *n* = 6 Control[2]: *n* = 7 **Background** C57BL6J **Age (mon)** 3 **Gender** Intervention: 3 Male 4 Female Control[1]: *n* = 3 Male 4 Female Control[2]: 3 Male 3 Female	<u>Clinical</u> N/R <u>Biochemical</u> Expression of growth factor - PGRN Expression of autophagy marker - LC3II	N/R	Trehalose increased expression of PGRN and LC3II in brain tissue of transgenic mice. Increased expression of PGRN is correlated with increased LC3II.	N/R	Trehalose activates growth factor in FTD through autophagy-lysosome modulators.

Rodríguez-Navarro et al. (2010) [8] Spain	Yes	**Intervention[1]:** Trehalose (1%, oral administration, 2x/wk, 2.5 mon; *4 mon for 6 mice) **Intervention[2]:** Trehalose (1%, oral administration, 2x/wk, 3 wk) **Control:** Vehicle (normal drinking water)	**PK$^{-/-}$/Tau^{VLW} transgenic mice [tau pathology with parkinsonism]** Intervention[1]: *n* = 18 - 4 mon: *n* = 10 Intervention[2]: *n* = 13 Control: *n* = 18 - 4 mon: *n* = 10 **Background** N/R **Age (mon)** 3 *14 (Intervention[2]) **Gender** N/R	<u>Clinical</u> Actimeter - Motor & anxiety-related activity Y-maze - STM, motor activity, and anxiety Stride length - Gait <u>Biochemical</u> Level of plagues - Aβ - Tau - Phospho-tau Expression of astrogliosis - GFAP$^+$ Degree of dopaminergic deficit - TH$^+$ neurons - DAT - DA metabolites: DOPAC, HVA, 3-MT - NA - NA metabolites: MHPG - 5-HT - 5-HT metabolites: 5-HIAA - Pre-&post-synaptic DA markers Expression of autophagy marker	Trehalose improved STM, motor activity, anxiety, and gait in transgenic mice after 2.5 mon and maintained after 4 mon *Older mice: Trehalose improved motor and anxiety-related activities in transgenic mice after 3 wk.	Trehalose decreased level of phospho-tau in striatum, limbic system (hippocampus), and hemispheres after 2.5 and 4 mon; decreased level of total tau in limbic system and striatum of transgenic mice after 2.5 mon. Trehalose decreased expression of GFAP$^+$ in the hippocampus, cerebral cortex, and striatum of transgenic mice; effect is greater in 2.5 mon than 4 mon. Trehalose increased numbers of TH$^+$ neurons, DAT, DOPAC, HVA, pre- & post-synaptic DA protein markers in striatum of transgenic mice after 2.5 mon; effect was not observed after 4 mon. Trehalose decreased level of p62/SQSTM1 in striatum of transgenic mice after 2.5 and 4 mon. Trehalose increased levels of LC3II, LC3II/LC3I ratio, and LC3II/β-actin ratio in	N/R	Trehalose improved motor and cognitive function in tauopathy by decreasing level of protein aggregates through activation of autophagy, inhibition of astrogliosis, and reduction of dopaminergic deficit.

	- p62/SQSTM1 - LC3II, LC3II/LC3I ratio, LC3II/β-actin ratio - Beclin-1 - Cytochrome c Oxidative stress - GSH Apoptosis - BCL2/BAX - Cleaved/total caspase-3	striatum of transgenic mice after 2.5 and 4 mon. Trehalose decreased cytochrome c in striatum of transgenic mice and maintained after 4 mon. Trehalose increased GSH in striatum of transgenic mice after 2.5 mon; effect not observed after 4 mon. Trehalose decreased BCL2/BAX and cleaved/total caspase-3 in striatum of transgenic mice after 2.5 mon but not 4 mon. *Older mice: Trehalose decreased insulin and Aβ, tau, and phosphor-tau in transgenic mice after 3 wk.

Schaeffer et al. (2012) [9] Schaeffer and Goedert (2012) [10] UK	N/R	**Intervention:** Trehalose (1%, oral administration, 2x/wk) **Control[1]:** Sucrose (2%, oral administration, 2x/wk) **Control[2]:** Vehicle (normal drinking water)	**P301S tau transgenic mice [tau pathology]** Intervention: *n* = 6 Control[1]: *n* = 6 Control[2]: *n* = 6 **WT** Intervention: *n* = 6 Control[1]: *n* = 6 Control[2]: *n* = 6 **Background** C57BL/6 **Age (mon)** 3 – 4 **Gender** N/R	<u>Clinical</u> N/R <u>Biochemical</u> Expression of autophagy - Conversion of LC3I to LC3II - p62/SQSTM1 Level of plagues - Tau - Phospho-tau - AT100 tau Neuronal survival - Number of neurons	N/R	Co-localization of p62/SQSTM1, LC3 and AT100 tau in transgenic mice. Trehalose increased conversion of LC3I to LC3II in the brain of transgenic and WT mice; not observed in spinal cord. Trehalose decreased p62/SQSTM1 in transgenic and WT mice; no difference observed between transgenic and WT mice. Trehalose decreased insoluble but not soluble tau in the brain of transgenic and WT mice; not observed in spinal cord. Trehalose decreased AT100 tau in positive cells in layer I-III of cerebral cortex and brainstem (pontine nucleus) in transgenic mice. Trehalose increased number of neurons in layers I-III of cerebral cortex and pontine nucleus of transgenic and WT mice.	N/R	Trehalose improved neuronal survival by decreasing level of protein aggregates in tauopathy through activation of autophagy.

			Number of neurons did not differ between trehalose-treated transgenic and untreated WT mice.	

[+]: Positive; 3-MA: 3-methyladenine; 3-MT: 3-methoxytyramine;5-HIAA: 5-Hydroxyindoleacetic acid; 5-HT: serotonin; AD: Alzheimer's disease; APP: Amyloid precursor protein; Aβ: beta amyloid; BACE1: β-site APP cleaving enzyme 1; BCL2/BAX: B-cell lymphoma 2-associated X protein; CHIP: C-terminus of Hsc70-interacting protein; CTF: c-terminal fragment; DA: dopamine; DAT: dopaminergic transporter; DOPAC: 3,4-dihydroxyphenylacetic acid; FTD: frontotemporal dementia; GFAP: glial fibrillary acidic protein; GSH: glutathione; HSP70: heat

shock protein 70; HVA: homovanillic acid; LC3: microtubule-associated protein 1A/1B-light chain 3; MHPG: 3-methoxy-4-hydroxyphenylglycol; N/R: not reported; NA: noradrenaline; PGRN: progranulin; PS1: presenilin 1; RCT: randomised controlled trial; SQSTM1: sequestosome 1; STM: short-term memory; TH: tyroxine-hydroxylase; TUNEL: terminal deoxynucleotidyl transferase (TdT) dUTP nick-end labeling; WT: wild-type.